

25.2 Cosmology

Question Paper

Course	CIEA Level Physics
Section	25. Astronomy & Cosmology
Topic	25.2 Cosmology
Difficulty	Easy

Time allowed: 40

Score: /26

Percentage: /100

Question 1a

State Hubble's law.

[1 mark]

Question 1b

Explain how the Hubble constant can be used to determine an age for the universe.

[1 mark]

Question 1c

The Hubble constant is estimated to be $2.2 \times 10^{-18} \text{ s}^{-1}$

(c)

Calculate the age of the Universe in years.

$$1 \text{ year} = 3.15 \times 10^7 \text{ s}$$

[3 marks]

Question 1d

Hubble's law and the idea of the expanding Universe support the Big Bang theory of the origin of the Universe. Table 1.1 contains statements about this theory.

(d)

Place ticks (✓) next to the statements that are correct.

Table 1.1

Statement	Correct (✓)
The universe began 14 million years ago	
There was a giant explosion known as the big bang	
This caused the universe to expand from a single point	
It heated as it expanded to form the universe today	
Each point in the universe expands away from the others	
So, the further away galaxies are the slower they are moving	

[3 marks]

Question 2a

State what is meant by redshift.

[2 marks]

Question 2b

Explain how cosmologists are able to determine that light from a distant star has undergone redshift.

[2 marks]

Question 2c

The Big Bang theory attempts to explain the origin of the Universe.

(c)

Describe the main ideas of the Big Bang theory.

[3 marks]

Question 2d

Explain why the redshift of galaxies is considered to be evidence for the Big Bang theory.

[2 marks]

Question 3a

The visible part of the electromagnetic spectrum from a star includes a dark line. This line is at a specific wavelength.

Fig 1.1 shows the position of the dark line in the spectrum from the Sun and in the spectrum from two different galaxies, galaxy A and galaxy B.

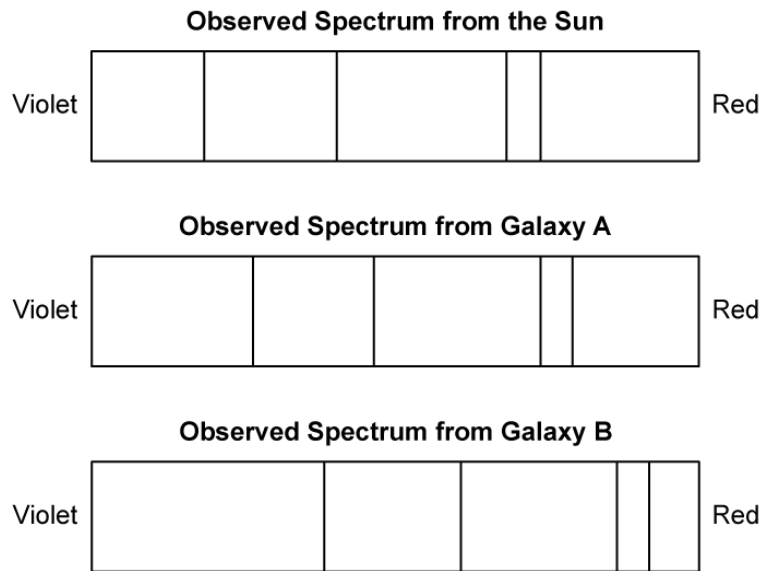


Fig. 1.1

(a)

Explain what the spectrum 'shifts' of the dark lines tell us about the direction of the galaxies.

[2 marks]

Question 3b

Using Fig. 1.1, state which galaxy is

(i)

moving faster,

[1]

(ii)

closest to the Earth.

[1]

[2 marks]

Question 3c

Galaxy A is at a distance of 150×10^9 km away from the Earth.

(c)

Calculate the recession velocity of galaxy A.

Hubble constant = $2.2 \times 10^{-18} \text{ s}^{-1}$

[3 marks]

Question 3d

A galaxy C is twice as far as galaxy A.

(d)

Determine the recession velocity of galaxy C.

[2 marks]

Model Answers: Easy

1a

(a) Hubble's law states:

EITHER

- The (recessional) speed of a galaxy is (directly) proportional to its distance from the Earth; [1 mark]

OR

- $H_0 = \frac{v}{d}$, where v = speed of recession of galaxy (from observer), d = distance of galaxy (away from observer); [1 mark]

[Total: 1 mark]

1b

(b) The Hubble constant can be used to determine an age for the universe from:

- $H_0 = \frac{v}{d}$, the Hubble constant H_0 has units of s^{-1}
- (Therefore) the age of the universe = $\frac{1}{H_0}$ [1 mark]

[Total: 1 mark]

Notice $\frac{1}{H_0}$ will have units of **s** (seconds) which can then be converted into years for the age of the universe.

1c

(c) Calculate the age of the universe in years.

List the known quantities:

- The Hubble constant, $H_0 = 2.2 \times 10^{-18} \text{ s}^{-1}$
- 1 year = $3.15 \times 10^7 \text{ s}$

Substitute the values into the age of the universe equation:

- Age of the universe = $\frac{1}{H_0}$
- Age of the universe = $\frac{1}{2.2 \times 10^{-18}} = 4.54 \times 10^{17} \text{ s}$ [1 mark]

Convert the age from seconds into years:

- Age of the universe = $\frac{4.54 \times 10^{17}}{3.15 \times 10^7}$ [1 mark]
- Age of the universe = $1.4 \times 10^{10} \text{ years}$ [1 mark]

[Total: 3 marks]

The age of the universe will always come out as around 14 billion years ($14 \times 10^9 \text{ years}$), so if your answer in the exam is ever wildly off from this you know that something has gone wrong in your calculation (most likely some powers or putting numbers in your calculator incorrectly!).

You can also remember this value from the intro theme tune to 'The Big Bang Theory' American sitcom:

*'Our whole universe was in a hot, dense state
Then nearly fourteen billion years ago expansion started, wait'*

1d

(d) The correct ticks (✓) are as follows:

Statement	Correct (✓)
The universe began 14 million years ago	
There was a giant explosion known as the big	

bang	✓
This caused the universe to expand from a single point	✓
It heated as it expanded to form the universe today	
Each point in the universe expands away from the others	✓
So, the further away galaxies are the slower they are moving	

- *One mark for each correct tick*

[Total: 3 marks]

- The universe began 14 **billion** and not 14 million years ago
- The universe has **cooled** and not heated as it has expanded
- The further away galaxies are, the **faster** they are moving

2a

(a) Redshift is:

- The apparent wavelength of light is observed to be greater than its known / measured value; [1 mark]
- (Due to) an object / star / galaxy moving away (from an observer) [1 mark]

[Total: 2 marks]

2b

(b) Cosmologists are able to determine that light from a distant star has undergone redshift by:

- Examining the (lines in the) spectrum (of light from the star); [1 mark]
- And comparing the lines with its known spectrum (of light measured on Earth); [1 mark]

[Total: 2 marks]

2c

(c) The main ideas of the Big Bang theory are:

- The Universe started with an explosion / Big Bang; [1 mark]
- All matter and energy was concentrated at a (single) point; [1 mark]
- The explosion caused the universe to expand from this (single) point; [1 mark]

[Total: 3 marks]

2d

(d) The redshift of galaxies is considered to be evidence for the Big Bang theory because

- If the Universe is expanding, then it must have been closer together in the past; [1 mark]
- The expansion of the Universe can be extrapolated back to its point of origin (known as the Big Bang); [1 mark]

[Total: 2 marks]

If the Universe is expanding at a measurable rate, then it makes sense that if we were able to rewind this expansion we would find a point in time where the Universe existed at a single point in time and space (called a singularity).

This is the point at which the Big Bang is thought to have occurred. By doing this, scientists have been able to estimate the age of the Universe to be about 13.7 billion years old.

3a

(a) The spectrum 'shifts' of the dark lines tell us that the galaxies are:

- Moving away from us / from Earth; [1 mark]
- Because the lines are shifted towards the red end of the spectrum / they have been redshifted; [1 mark]

[Total: 2 marks]

becauseL

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- Moving away from us / from Earth; [1 mark]
- Because the lines are shifted towards the red end of the spectrum / they have been redshifted; [1 mark]

[Total: 2 marks]

3b

(b)

(i) The galaxy moving faster is:

- Galaxy B; [1 mark]

(ii) The galaxy that is closer to the Earth is:

- Galaxy A; [1 mark]

[Total: 2 marks]

The greater the redshift, the faster a galaxy is moving away from us. Hubble's law

states that the further away a galaxy is, the faster it is moving away from us.

3c

(c) Calculate the recession velocity of Galaxy A:

List the known quantities:

- Distance, $d = 150 \times 10^9 \text{ km}$
- Hubble constant = $2.2 \times 10^{-18} \text{ s}^{-1}$

Use Hubble's law to calculate recession velocity:

- Hubble's law: $v = H_0 d$ [1 mark]
- $v = (2.2 \times 10^{-18}) \times (150\,000 \times 10^9)$ [1 mark]
- $v = 3.3 \times 10^{-4} \text{ m s}^{-1}$ [1 mark]

[Total: 3 marks]

3d

(d) Determine the recession velocity of galaxy C:

List the known quantities:

- Recession velocity of Galaxy A = $3.3 \times 10^{-4} \text{ m s}^{-1}$ (from (c))

Calculate the recession velocity of galaxy C:

- Hubble's law: $v \propto d$
- Galaxy C velocity = $2 \times (3.3 \times 10^{-4})$ [1 mark]
- Galaxy C velocity = $6.6 \times 10^{-4} \text{ m s}^{-1}$ [1 mark]

[Total: 2 marks]

Since the Hubble constant is the same, if a galaxy is twice as far, it must be travelling at twice the speed. Speed and distance are directly proportional to each other according to Hubble's law.

